

Physical Chemistry (I)

Lecture 23

기말고사 안내 및 복습



기말고사 안내

- 시간: 6월 14일(수) 오후 1:30~2:30
- 장소: 화학관 414호(홀수 학번), 315호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

시험 범위

- 8장 "단순 혼합물"
- 9장 "화학 평형"
- 10장 "전기화학"

시험 문제

총 여섯 문제

- 과제 문제, 기출 문제 혹은 그 변형(3문제)
 - 유도라면 중간 단계를 주지 않을 수도 있음
- 개념 설명 및 공식 유도(2문제)
 - 강의 자료 기준
- 보너스 문제(1문제)



Chapter 8

- Phase
- Chemical potential
- Gibbs-Duhem eq.

Multi-component system

Gas mixture

- Dalton's law
- Perfect gas
- Real gas

Liquid mixture

Ideal solution

- Raoult's law
$$p_A = x_A p_A^*$$
- Chemical potential
$$\mu_A = \mu_A^* + RT \ln x_A$$
- $\Delta_{\text{mix}}G = -T\Delta_{\text{mix}}S$

Ideal-dilute solution

- Henry's law
$$p_B = x_B K_B$$
- Chemical potential
$$\mu_B = \mu_B^\ominus + RT \ln x_B$$

Real solution

- Regular solution
- Bragg-Williams model
- Excess function
- Activity
$$\mu_A = \mu_A^* + RT \ln a_A$$
$$\mu_B = \mu_B^\ominus + RT \ln a_B$$

- L-G phase diagram
- L-L phase diagram
- L-S phase diagram

Colligative properties

Concepts and Derivations in Chpt 8

- Extension to multi-component systems
 - [C] Thermodynamic potentials
 - [C, D] Partial molar quantities
 - [C, D] Chemical potential
 - [D] Gibbs-Duhem equation
- [C] Dalton's law
- [D] Mixing of perfect gases
- [C] Real gases: fugacity and fugacity coefficient

Concepts and Derivations in Chpt 8

- [C] Mixing of liquids: liquid-gas equilibrium
- [C] Ideal solutions
 - [C] Raoult's law and ideal solutions
 - [D] Chemical potential
 - [D] Thermodynamics of mixing
 - [D] Liquid and vapor compositions
- [C] Ideal-dilute solutions
 - [C] Henry's law and ideal-dilute solutions
 - [C] (Idealized) chemical potentials of solute and solvent



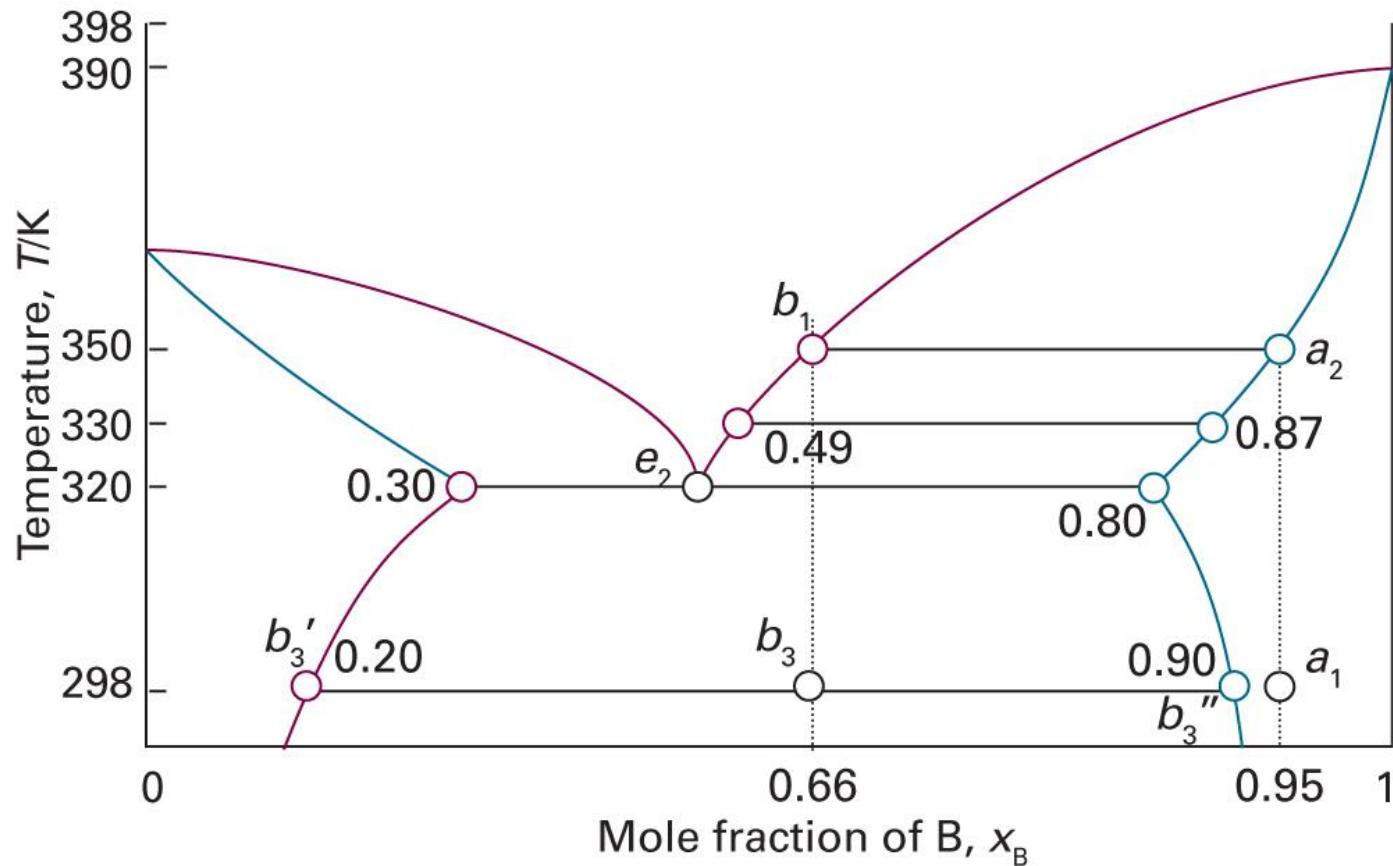
Concepts and Derivations in Chpt 8

- [C] Regular solutions
- [C] Bragg-Williams model
 - [D] Entropy of mixing
 - [D] Energy of mixing
 - [D] Helmholtz energy of mixing
- [C, D] Excess function
- [C, D] Activity and activity coefficient
 - [D] Activity of Bragg-Williams model

Concepts and Derivations in Chpt 8

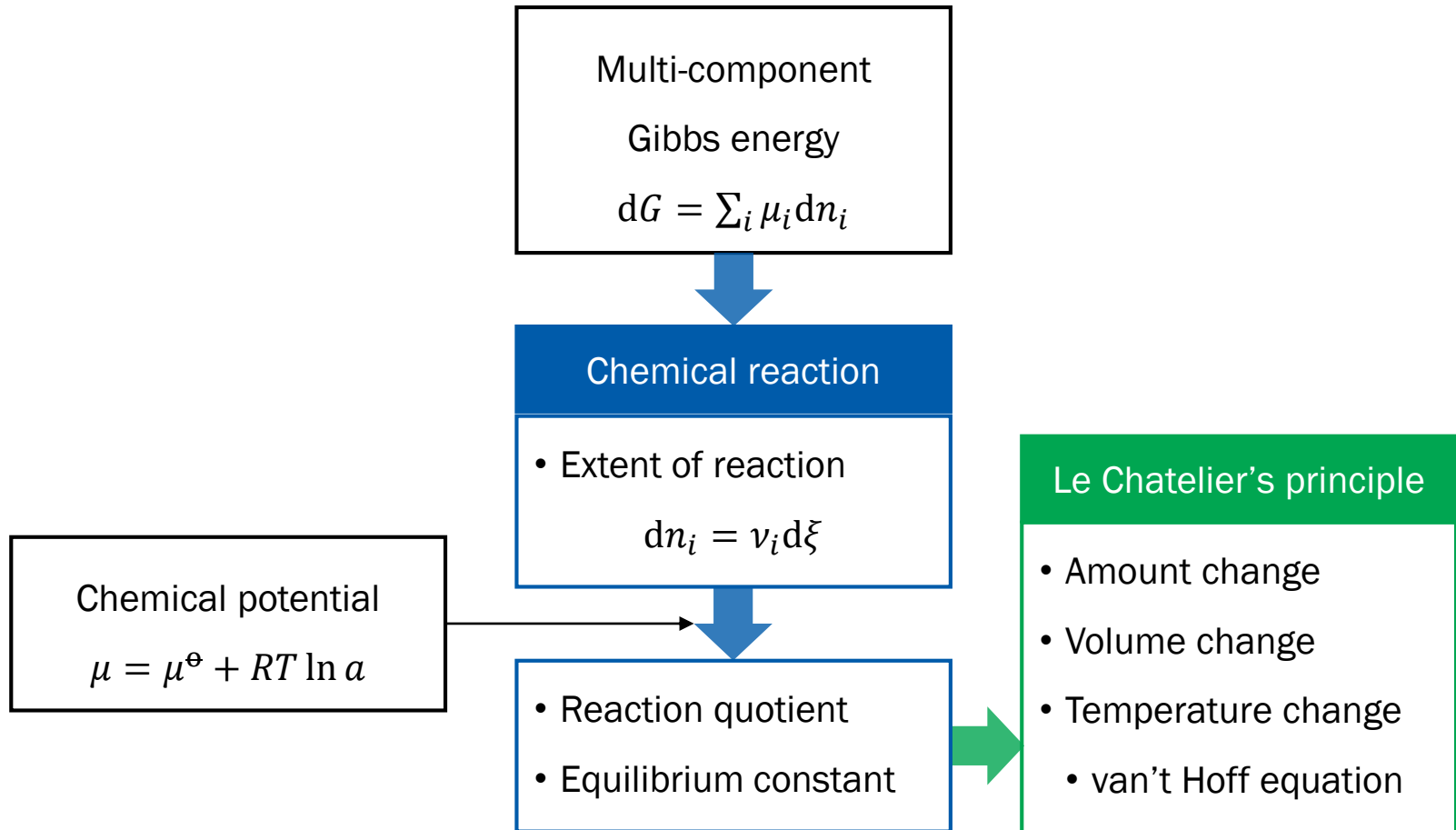
- [C] Phase diagrams
 - [C] Ideal solutions
 - [C, D] Real solutions: azeotropes and phase separation
 - [C] Liquid-solid mixtures
- [C, D] Colligative properties
 - Lowering of vapor pressure
 - Elevation of T_b and depression of T_f
 - Osmotic pressure
 - van't Hoff factor

Interpretation of a Phase Diagram



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.22)

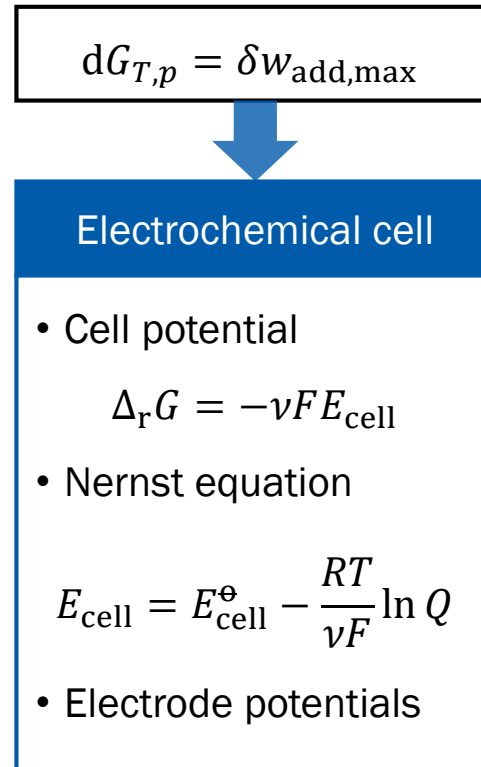
Chapter 9



Concepts and Derivations in Chpt 9

- [C, D] Reaction quotient and equilibrium constant
- [C, D] Approximation to activity
- [D] Gaseous reactions: K_p and K_c
- [C, D] Le Chatelier's principle
 - Amount change
 - Volume change
 - Temperature change: van't Hoff equation

Chapter 10



Concepts and Derivations in Chpt 10

- [C] Electrochemical cells
- [D] Thermodynamics of cells
- [D] Nernst equation
- [D] Entropy and enthalpy of cell reactions
- [C, D] Electrode potentials